Name of Company	Krutanic
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Project	Minor – Network Packet Analysis
Tools	Wireshark, VirtualBox, Kali-Linux, Nmap, Metasploitable 2

Network Packet Analysis for Threat Detection Report

Objective: Analyze real-time network traffic to detect malicious patterns

- **Attacker & IP:** Kali-Linux – 192.168.56.102
- **Target & IP:** Metasploitable 2 – 192.168.56.101
- **Gateway & IP:** Windows 11 – 192.168.56.1

1. **Port scanning attempts:**
   - **Wireshark filter:** (tcp.flags.syn == 1 or tcp.flags.reset == 1) and ip.addr == 192.168.56.101
   - **Command used on Kali-Linux to initiate SYN scan:** nmap -sS <Target_IP>
     Attacker was observed systematically probing the target for open ports.

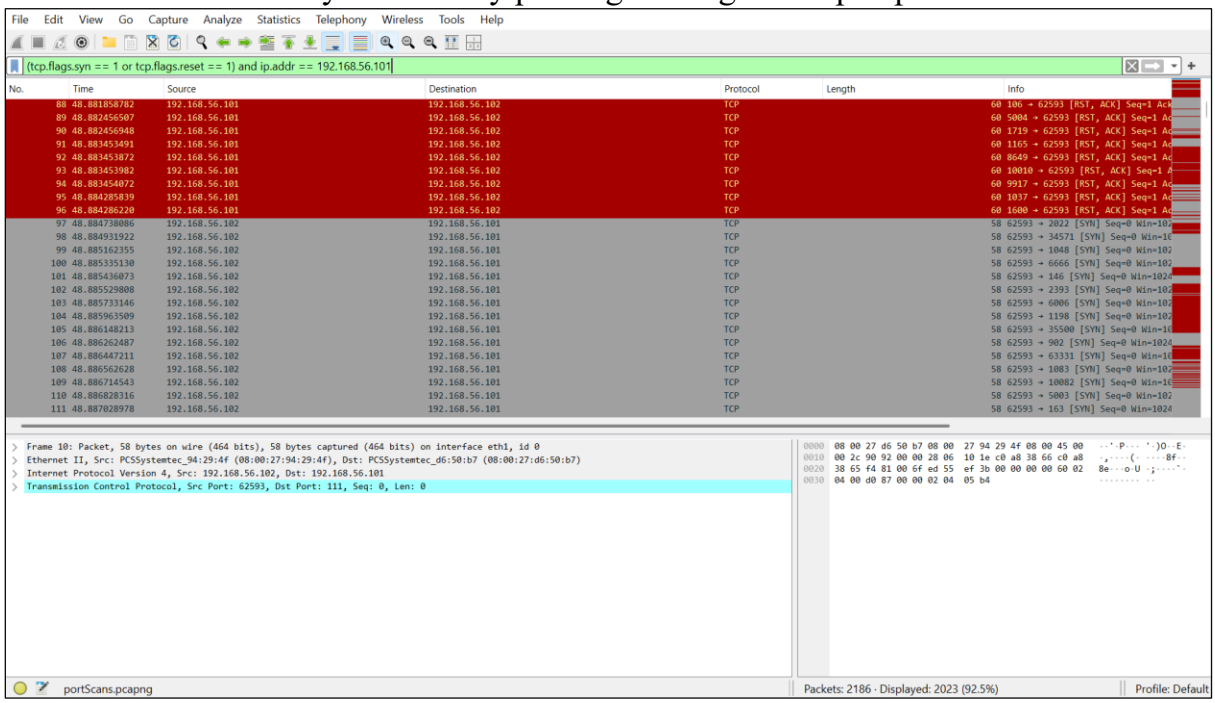

Figure I. Port Scanning Attempts with TCP SYN scan (grey)

Findings:

- In a port scan, the attacker sends a rapid stream of TCP SYN packets to the target aimed at different ports to check whether any ports are open. This way the attacker can gain access to the target through those open ports.
- The first step in a normal TCP 3-way handshake is to send SYN request for a connection.
- But the above packet capture image shows the attacker with the IP 192.168.56.102 is rapidly sending automated TCP SYN packets to the target with IP 192.168.56.101 with seq=0, len=0 which indicates there is no data payload, displaying the connection is just to check if any ports are open (as shown in grey color coded packets).
- The red packets (88-96) are from the target [RST, ACK] which suggests target saying the ports are closed.

Recommendations:

- Place sensitive servers on private subnets or on isolated VLANs.
- Close unused ports.
- Allow only trusted IPs wherever possible.
- Limit how many SYN packets a single IP can send per second.

- Tools like Snort, Suricata can detect and block port scanning in real-time.
- Block source IP at firewall.
- Close exposed ports discovered during scan.
- Patch vulnerabilities in case further account access gained.
- Investigate further for login access to user account and privilege escalation.

2. **ARP Spoofing/ Man-In-The-Middle Attack (MITM):**
   Commands to generate a bidirectional MITM attack in Kali-Linux:
   Terminal 1: sudo arpspoof -i eth1 -t 192.168.56.101 192.168.56.1
   Terminal 2: sudo arpspoof -i eth1 -t 192.168.56.1 192.168.56.101
   In the Terminal 1 command,
   - 192.168.56.1(spoofed gateway IP) tells 192.168.56.101(Metasploitable or target IP) that "I am the gateway, send all your traffic to me".
   - The target believes this and updates its ARP table which now saves the spoofed gateway MAC address as the real gateway.
   - All the traffic from the target is now going to the attacker's MAC address.

   In the Terminal 2 command,
   - The attacker now acts as the target and asks the real gateway to send all traffic to its IP.
   - The gateway updates its ARP table to the spoofed target's MAC.
   - So now all the traffic from actual gateway (192.168.56.1) goes to spoofed target (192.168.56.101).

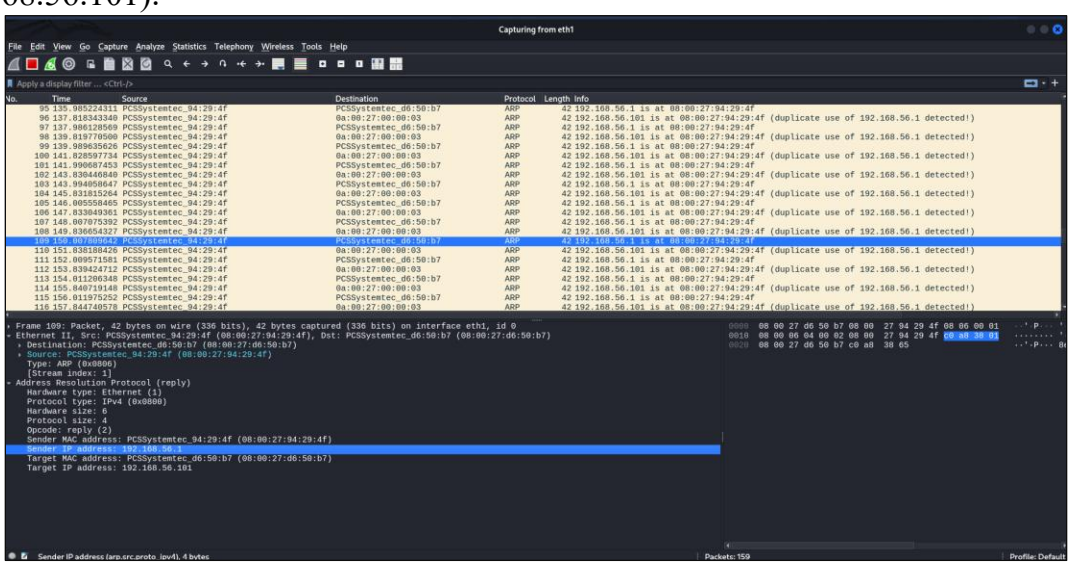

Figure II. ARP Spoofing

Findings:
- The ARP packet capture shows a high volume of ARP reply packets where a single MAC address (08:00:27:94:29:4f) repeatedly says that "IP address 192.168.56.1 is mine", which is usually assigned to the network gateway.
- But since it has been spoofed, this IP is now the attacker so all the traffic from 192.168.56.101 (target/victim IP) goes to attacker IP and the target updates its ARP table thinking this is the real gateway MAC address.
- Wireshark flags this as a "duplicate use" condition, indicating that multiple devices are trying to use same IP address.

Recommendations:

- Enforce HTTPS instead of HTTP for all web traffic.
- Use secure protocols like SSH instead of TLS.
- Manually configure MAC address for critical systems and servers.
- Enable Dynamic ARP Inspection on network switches to prevent unauthorized MAC to IP mapping.
- Block attacker MAC/IP on firewall.
- Check for unusual machine/system activity and isolate all those affected from the network.

3. Unusual DNS requests:

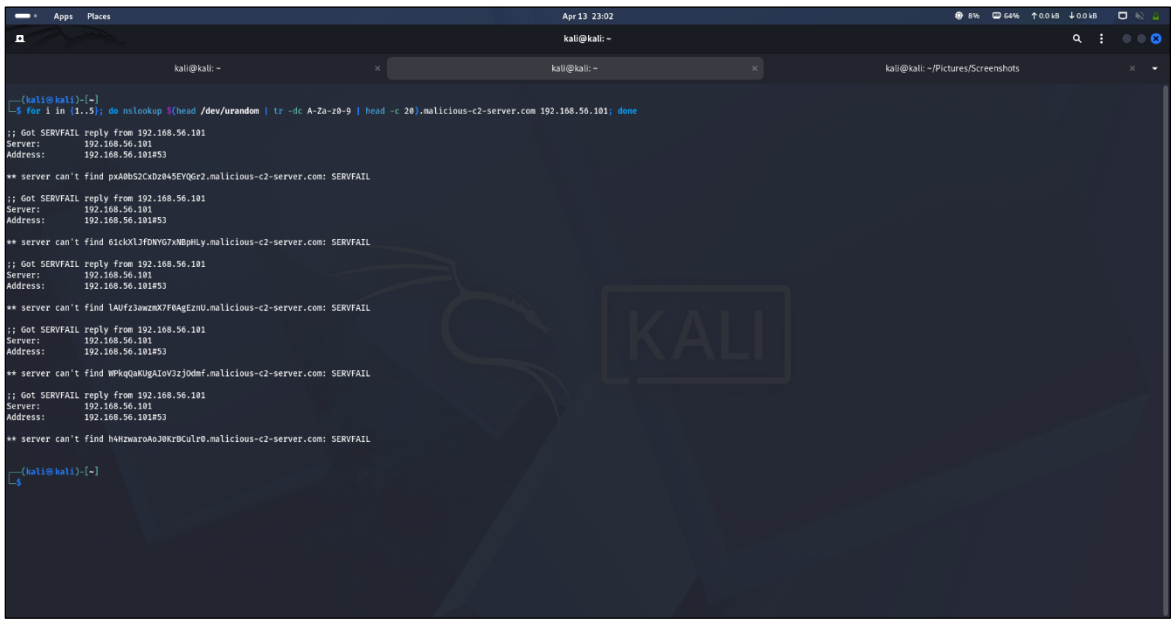

Figure III. Domain Generation Algorithm (DGA)

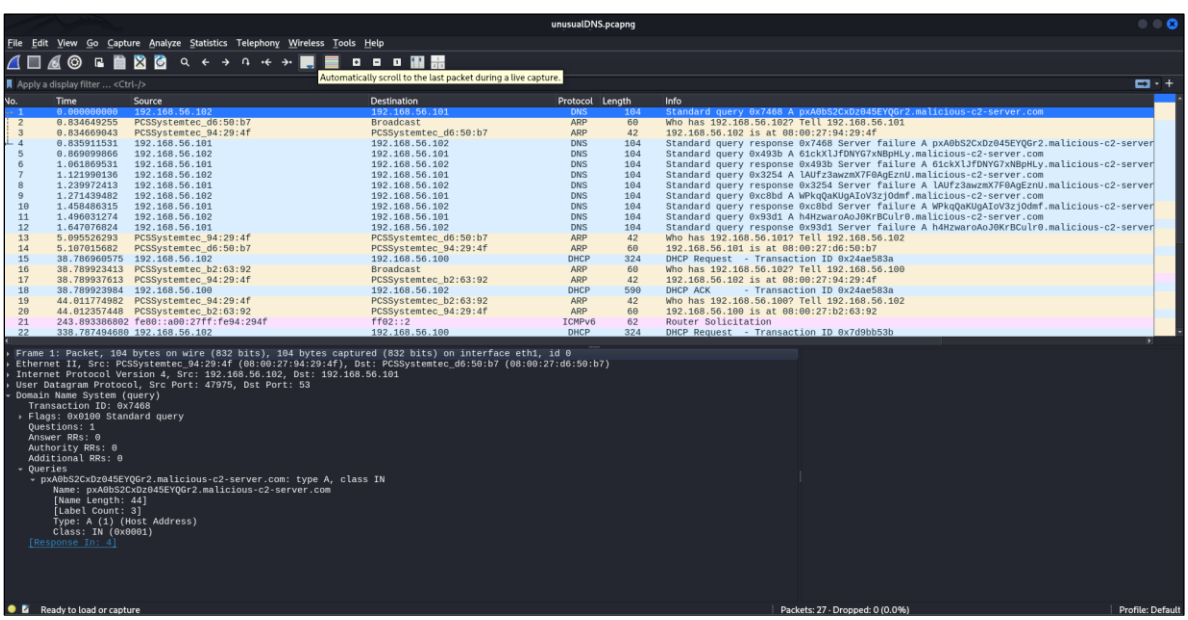

Figure IV. DGA Packet Capture

Findings:

- The figure III shows repeated DNS queries for randomly generated subdomains under a fixed domain all of which respond with SERVFAIL.
- This is consistent with domain generation algorithm activity.
- This is used by malware to detect Command and Control(C2) infrastructure.
- The figure IV shows the packet capture where the system with IP 192.168.56.102 acts as the client that send requests to locate a C2 infrastructure.
- Metasploitable with IP 192.168.56.101 acts as the DNS server, and send SERVFAIL responses to the suspicious client requests.

Recommendations:

- Implement Domain Name System Security Extensions (DNSSEC).
- Use firewalls to allow only known and trusted IPs and domains.
- Use DNS over HTTPS and DNS over TLS to protect queries from interception and modification.
- Actively audit DNS logs for high volume of requests for subdomains or unusual DNS record types like TXT, MX.